Машинное обучение в PostgreSQL pg_ml



Data Fest 2024

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2024

Концепция Большие данные

- Предложена в 2008 Клиффорд Линч
- Не данные идут к коду, а код идет к данным.

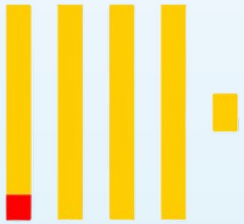
Немного истории

- 1992 PL/SQL (Oracle, Db2)
- 1992 T-SQL (SQL Server)
- 1996 PSQL (Postgres SQL)
- 1999 MySQL 3.23
- 2006-2016 Google BigQuery
- 2006-2008 Hadoop
- 2010-е начало активного внедрения ML в бизнес


amazon
REDSHIFT




VERTICA






Microsoft®
SQL Server®

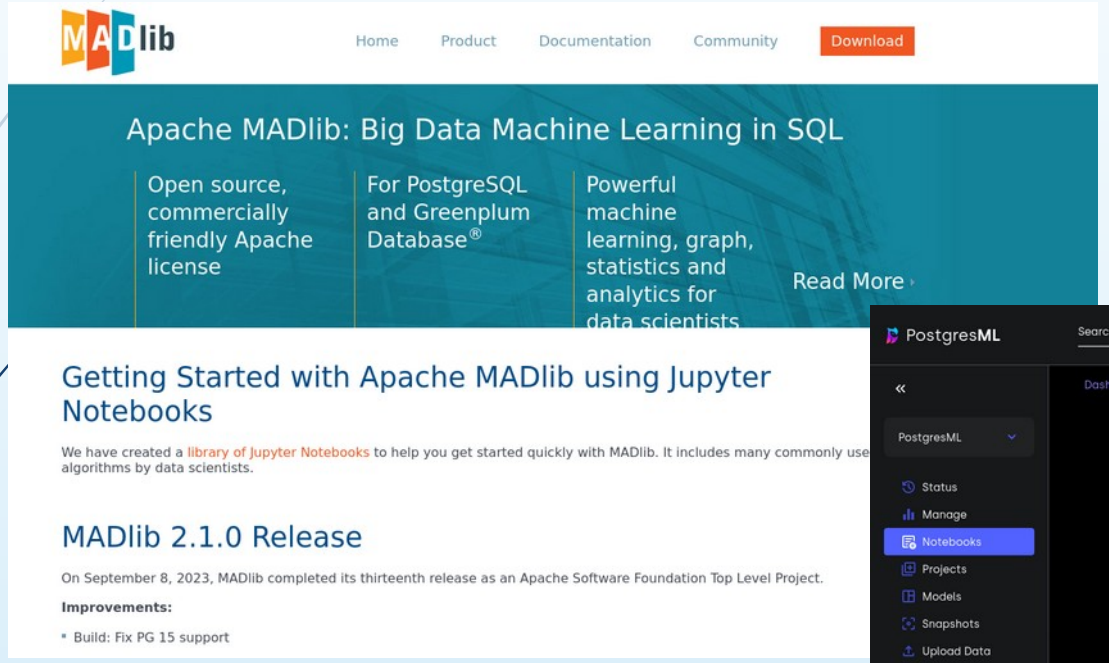

ARENADATA


ORACLE®
DATABASE

Почему БД?

- Табличные данные
- Отсутствие промежуточных скриптов
- Все результаты получаются в запросах

Проекты ML в PostgreSQL

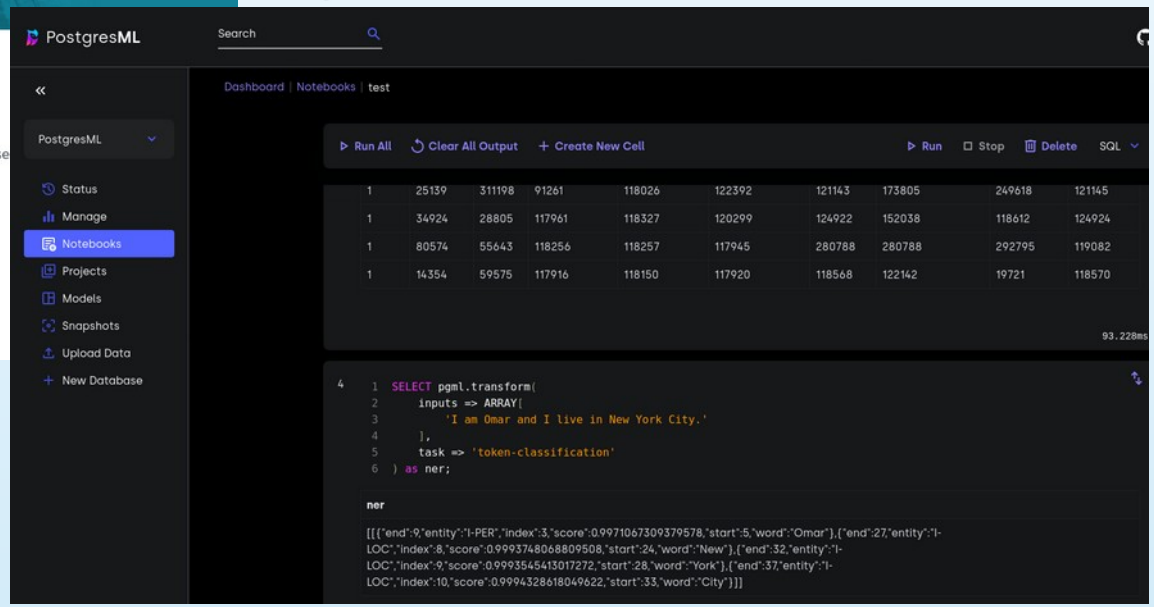




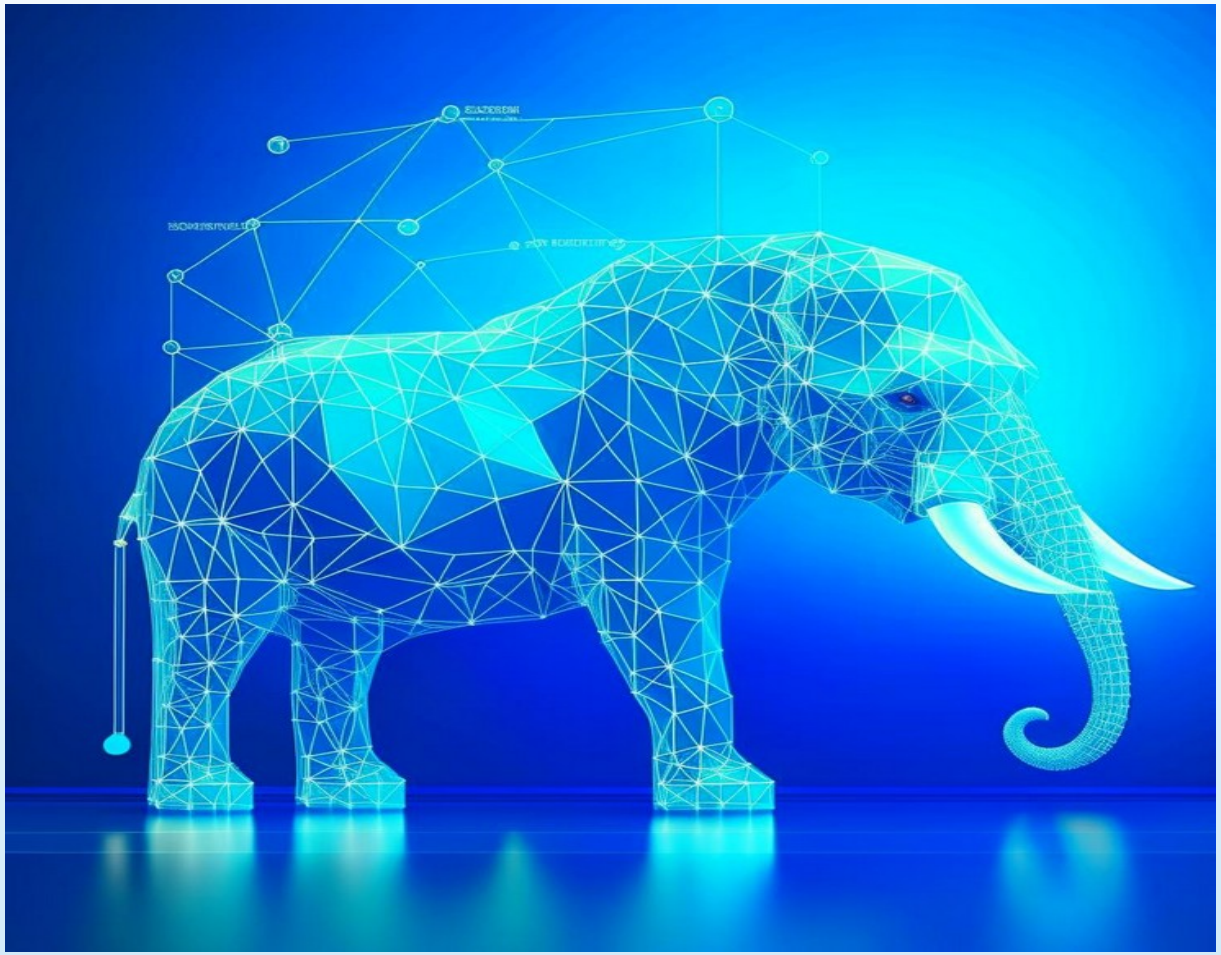
Google's AutoML



Google's AutoML



pg_ml



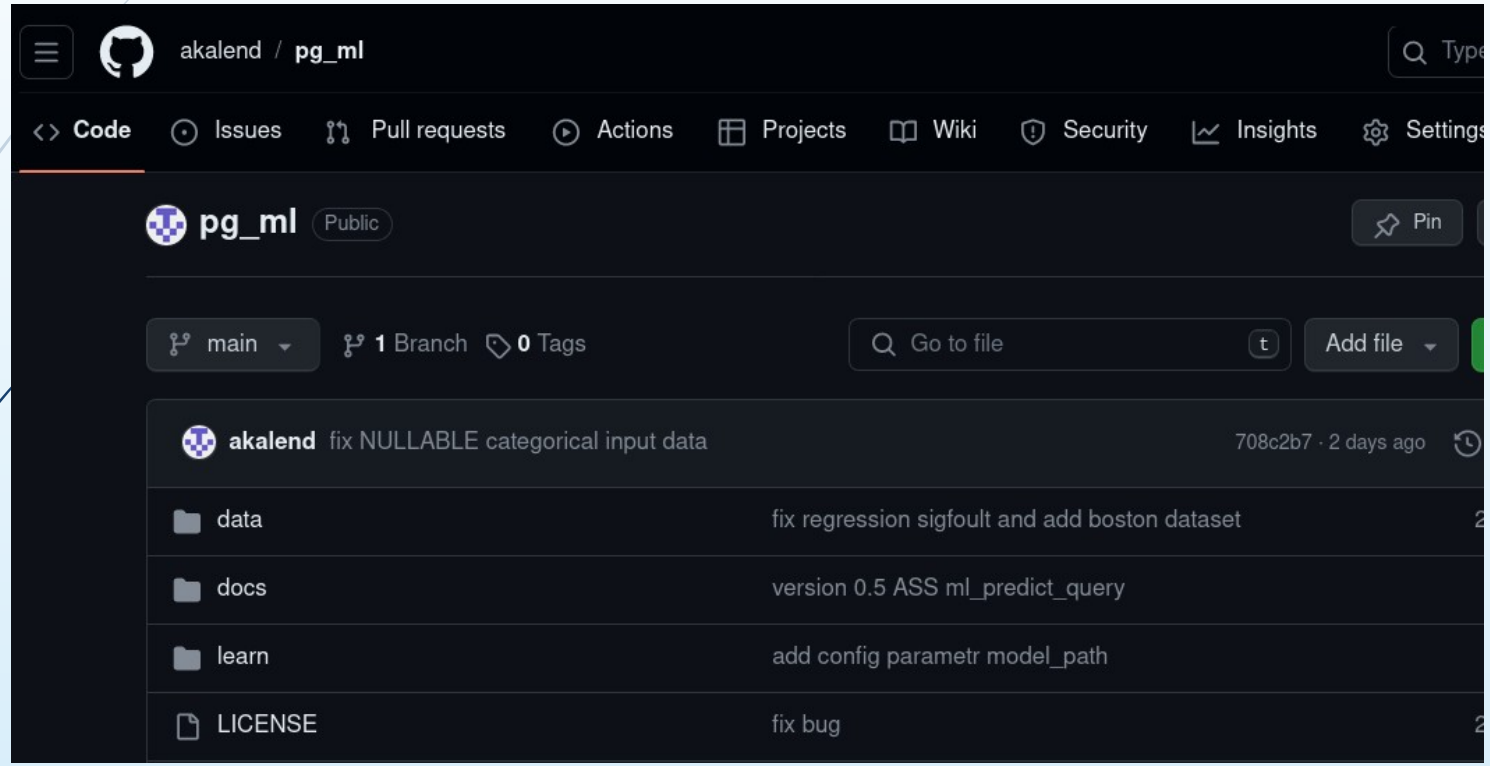
Основан на разработках Яндекс

CatBoost: https://catboost.ai/
libcatboostmodel.so



реализовано только предсказание
(ограничение libcatboostmodel)

http://github.com/akalend/pg_ml

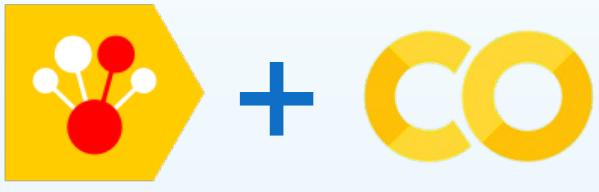


Доступные модели:

- Binary Classification
- Multi Classification
- Regression
- Ranking (проверка)
- Text classification (перспектива)

ML процес

1. Тенировка модели
2. Сохранение на сервер БД
3. предсказание



```
model.save_model('astra.cbm')
!ls
```

```
astra.cbm  astra.json  catboost_i
```

```
adult=# SELECT ml_predict ('astra3.cbm','astra3');
WARNING:  field run_id not used
WARNING:  field field_id not used
WARNING:  field spec_obj_id not used
WARNING:  field predict not used
      ml_predict
-----------------------
 public.astra3_predict
(1 row)
```

Соответствие имен полей модели и таблицы в БД



ID, Id, id

FieldID, Field_ID

Import-port

Port, PORT

<=>



id

field_id

import_port

port

Соответствие имен полей модели и таблицы в БД

```
[14] df = pd.read_csv('star_classification.csv')
```

```
[16] for it in df.columns:
       print(it)

     obj_ID
     alpha
     delta
     u
     g
     r
     i
     z
     run_ID
     rerun_ID
     cam_col
     field_ID
     spec_obj_ID
     class
     redshift
     plate
     MJD
     fiber_ID
```

<=>

```
adult=# \d astra3
                    Table "public.astra3
   Column    |        Type        | Collation
-------------+--------------------+-----------
 alpha       | double precision   |
 delta       | double precision   |
 u           | double precision   |
 g           | double precision   |
 r           | double precision   |
 i           | double precision   |
 z           | double precision   |
 run_id      | bigint             |
 cam_col     | bigint             |
 field_id    | bigint             |
 spec_obj_id | double precision   |
 redshift    | double precision   |
 plate       | bigint             |
 mjd         | bigint             |
 fiber_id    | bigint             |

adult=#
```

Получение результата:

- В виде таблицы
- Как набор записей из таблицы (представления)
- Как набор записей из запроса

Результ: таблица

Исходая таблица

модель

```
adult=# SELECT * from  ml_predict_table('astra3.cbm','astra3');
   ml_predict_table
-----------------------
 public.astra3_predict
(1 row)
```

Создается новая таблица

postgres@notebook-sasha: /usr/local/pgsql

```
adult=# select * from astra3_predict;
```

row	alpha	delta	u	g	r	i	z	run_id	cam_col	field_id	spec_obj_id	redshift	plate	mjd	fiber_id	predict	class
1	16.9568897845004	3.64613008870454	23.33542	21.95143	20.48149	19.603	19.13094	7712	6	442	4.855016555329904e+18	0.5062369	4312	55511	495	0.98686	GALAXY
2	240.063240247767	6.13413059813973	17.86033	16.79228	16.43001	16.30923	16.25873	3894	1	243	2.448928032270870 5e+18	0.0003448142	2175	54612	348	).990419	STAR
3	30.887222067625	1.18870964120799	18.18911	16.89469	16.42161	16.24627	16.18549	7717	1	536	8.255357438959835e+18	4.085216e-06	7332	56683	943	).997588	STAR
4	247.594400505002	10.8877797153666	24.99961	21.71203	21.47148	21.30532	21.29109	5323	1	134	4.577998722756271e+18	-0.0002914838	4066	55444	326	).997667	STAR
5	18.8964507920807	-5.26133022886992	23.76648	21.79737	20.69543	20.23403	19.97464	7881	3	148	8.91047176642785e+18	-0.0001361561	7914	57331	363	).996044	STAR
6	182.713733094955	51.3758050594777	22.44608	21.68444	20.24292	19.41423	19.08227	2830	1	411	7.516725588574623e+18	0.5026683	6676	56389	792	).984373	GALAXY
7	150.089423193165	39.4670880748061	18.96441	17.82906	17.31429	16.99891	16.85583	3560	4	278	1.526795641110423 6e+18	0.06366445	1356	53033	274	).996164	GALAXY
8	189.510984338851	58.7411197772507	21.37376	20.80187	20.84925	21.13449	20.34689	2243	1	353	7.696817897528907e+18	0.7936153	6836	56443	604	).957787	QSO
9	37.7138728560977	-0.525138228146508	20.77988	19.54618	19.16687	18.89438	18.64286	2700	2	117	1.755328312302921 7e+18	0.1060118	1559	53271	183	).993892	GALAXY
10	201.074980072746	28.7699058867715	25.05349	22.23362	20.8122	19.69488	19.28336	4649	3	120	7.306035245308205e+18	0.567082	6489	56329	257	).993856	GALAXY
11	151.83091832672	19.8108624669417	24.04443	22.48608	20.59701	19.50985	19.00457	5183	5	142	6.622787444780849e+18	0.5475619	5882	56029	888	).998885	GALAXY
12	164.364389178099	64.7877852553783	22.98745	23.07199	21.15229	19.97391	19.17876	1302	6	319	7.999579699214046e+18	0.7592601	7105	56740	221	0.99362	GALAXY
13	242.830624949049	52.6659057609914	23.03347	23.60839	21.79315	20.79881	19.74832	1412	4	375	9.081519495399626e+18	0.680221	8066	57544	39	).986432	GALAXY
14	22.9648494083052	0.876034964444952	25.62939	21.3913	19.98634	18.81294	18.13539	4263	6	269	1.213827896058079 2e+18	-1.408066e-05	1078	52643	392	0.99907	STAR
15	176.268553005359	33.432292927676	21.83964	19.87187	18.8392	18.3688	18.09703	4576	1	399	1.155301576648925 6e+19	0.1312984	10261	58462	570	).967815	GALAXY
16	227.751962030348	41.8324497606407	25.98832	23.93829	21.41312	20.34021	19.5577	3664	5	107	9.592801209263348e+18	0.5425886	8520	58191	487	).983403	GALAXY

Результат: набор данных (таблица)

Исходная таблица

Модель

Список категорийных полей

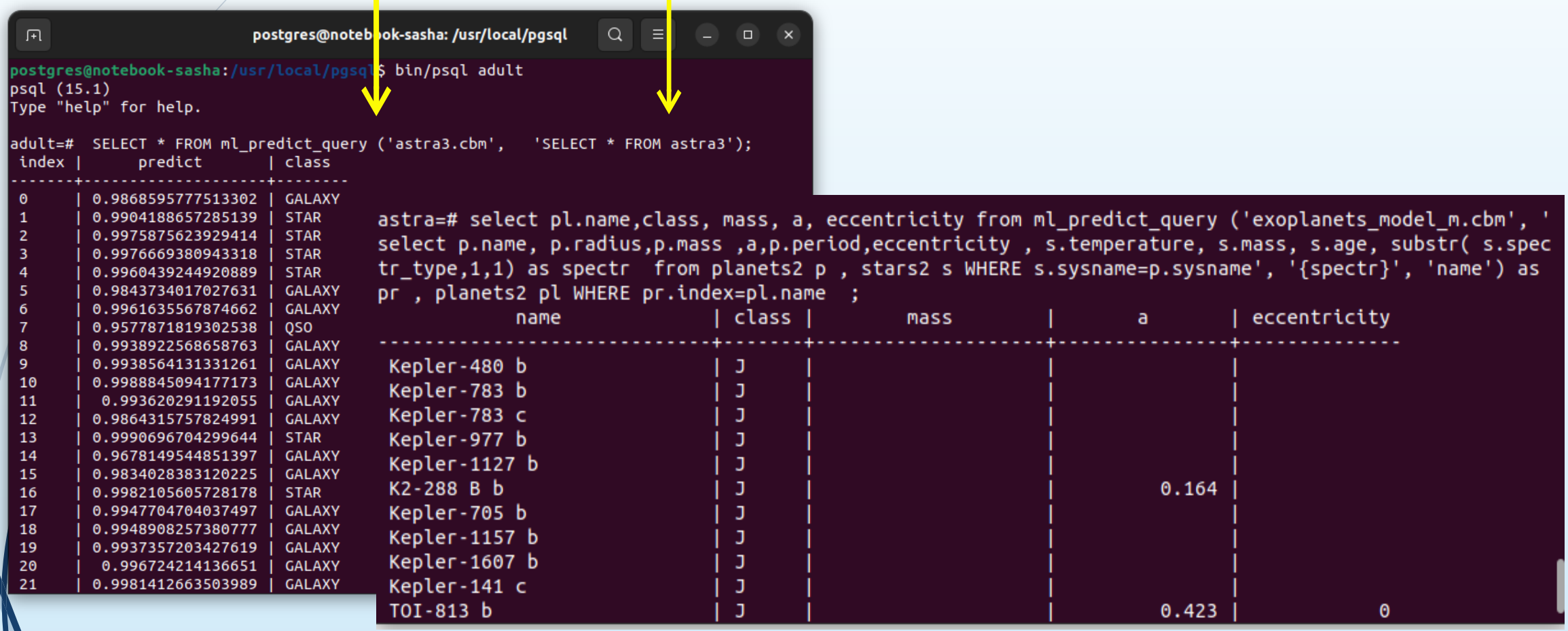
```
adult=#   SELECT * from ml_cat_predict ('titanic.cbm',
'titanic','{name,passenger_id,pclass,sex,sibsp,parch,ticke
t,cabin,embarked }');
 row_num |        predict        | class
---------+-----------------------+-------
       0 |   -1.7937342449233795 | 0
       1 |   -0.7958399022225136 | 0
       2 |    -2.392873216013247 | 0
       3 |    -1.942976624899004 | 0
       4 |  -0.41747860726736713 | 0
       5 |   -2.0608914711097546 | 0
       6 |    0.5914467057444344 | 1
       7 |   -1.0786526230973736 | 0
       8 |    0.6757411102494171 | 1
       9 |    -3.250956928980716 | 0
      10 |    -2.274725588104562 | 0
      11 |   -1.3228896775643357 | 0
      12 |      2.70931909246417 | 1
      13 |   -2.4233542239140187 | 0
```

Набор данных (четвертый параметр)

```
adult=# SELECT index as name, class FROM ml_predict('titanic.cbm', 'titanic', '{
name,passenger_id,pclass,sex,sibsp,parch,ticket,cabin,embarked }', 'name') LIMIT
 10;
                     name                      | class
-----------------------------------------------+-------
 Kelly, Mr. James                              | 0
 Wilkes, Mrs. James (Ellen Needs)              | 0
 Myles, Mr. Thomas Francis                     | 0
 Wirz, Mr. Albert                              | 0
 Hirvonen, Mrs. Alexander (Helga E Lindqvist) | 0
 Svensson, Mr. Johan Cervin                    | 0
 Connolly, Miss. Kate                          | 1
 Caldwell, Mr. Albert Francis                  | 0
 Abrahim, Mrs. Joseph (Sophie Halaut Easu)     | 1
 Davies, Mr. John Samuel                       | 0
(10 rows)
```

Результат: набор записей из запроса

модель **запрос**



```
postgres@notebook-sasha: /usr/local/pgsql
postgres@notebook-sasha:/usr/local/pgsql$ bin/psql adult
psql (15.1)
Type "help" for help.

adult=#  SELECT * FROM ml_predict_query ('astra3.cbm',   'SELECT * FROM astra3');
 index |      predict       | class
-------+--------------------+--------
 0     | 0.9868595777513302 | GALAXY
 1     | 0.9904188657285139 | STAR
 2     | 0.9975875623929414 | STAR
 3     | 0.9976669380943318 | STAR
 4     | 0.9960439244920889 | STAR
 5     | 0.9843734017027631 | GALAXY
 6     | 0.9961635567874662 | GALAXY
 7     | 0.9577871819302538 | QSO
 8     | 0.9938922568658763 | GALAXY
 9     | 0.9938564131331261 | GALAXY
 10    | 0.9988845094177173 | GALAXY
 11    |  0.993620291192055 | GALAXY
 12    | 0.9864315757824991 | GALAXY
 13    | 0.9990696704299644 | STAR
 14    | 0.9678149544851397 | GALAXY
 15    | 0.9834028383120225 | GALAXY
 16    | 0.9982105605728178 | STAR
 17    | 0.9947704704037497 | GALAXY
 18    | 0.9948908257380777 | GALAXY
 19    | 0.9937357203427619 | GALAXY
 20    |  0.996724214136651 | GALAXY
 21    | 0.9981412663503989 | GALAXY
```

```
astra=# select pl.name,class, mass, a, eccentricity from ml_predict_query ('exoplanets_model_m.cbm', '
select p.name, p.radius,p.mass ,a,p.period,eccentricity , s.temperature, s.mass, s.age, substr( s.spec
tr_type,1,1) as spectr  from planets2 p , stars2 s WHERE s.sysname=p.sysname', '{spectr}', 'name') as
pr , planets2 pl WHERE pr.index=pl.name  ;
            name             | class |        mass        |       a       | eccentricity
-----------------------------+-------+--------------------+---------------+--------------
 Kepler-480 b                | J     |                    |               |
 Kepler-783 b                | J     |                    |               |
 Kepler-783 c                | J     |                    |               |
 Kepler-977 b                | J     |                    |               |
 Kepler-1127 b               | J     |                    |               |
 K2-288 B b                  | J     |                    |         0.164 |
 Kepler-705 b                | J     |                    |               |
 Kepler-1157 b               | J     |                    |               |
 Kepler-1607 b               | J     |                    |               |
 Kepler-141 c                | J     |                    |               |
 TOI-813 b                   | J     |                    |         0.423 |             0
```

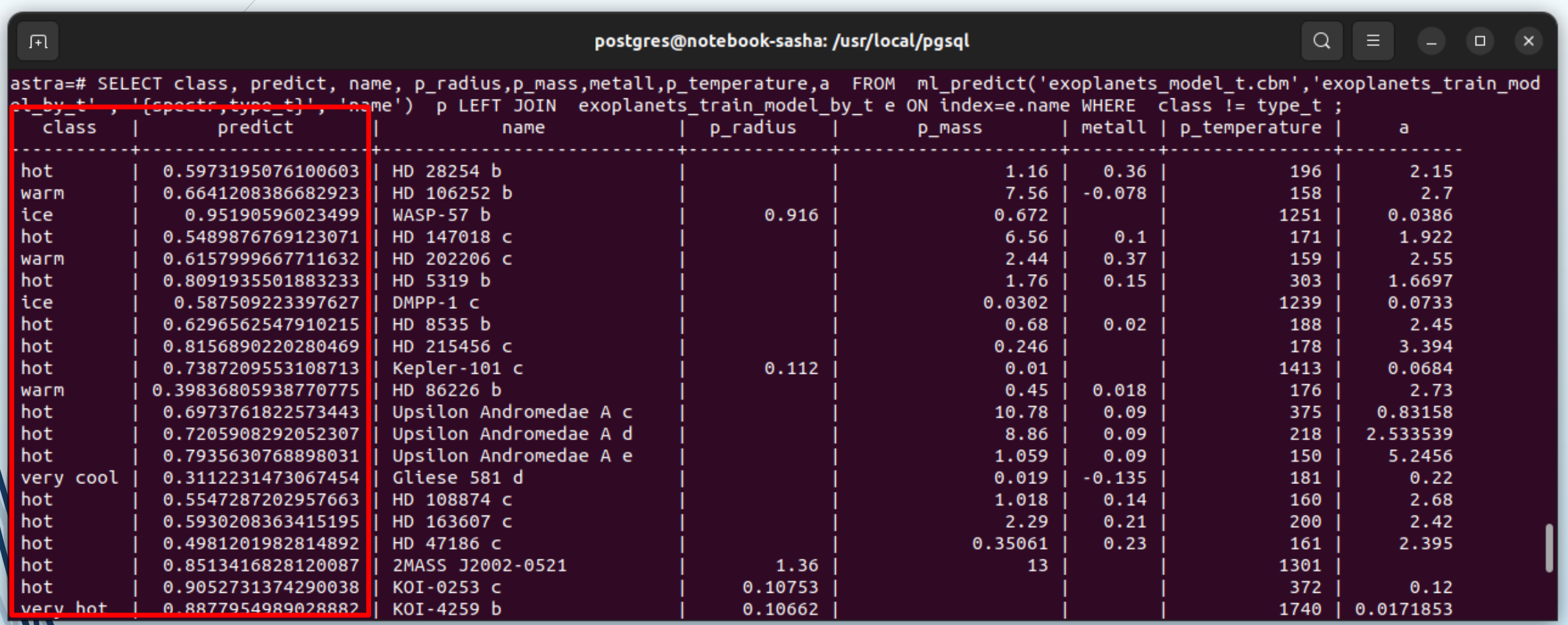
SELECT * FROM {table}_predict;

```
adult=# SELECT * from astra3_predict;
 row |       alpha       |       delta       |    u     |    g     |    r     |    i     |    z     | run_id | cam_col | field_id |      spec_obj_id      |   redshift    | plate | mjd | fiber_i | predict | class
-----+-------------------+-------------------+----------+----------+----------+----------+----------+--------+---------+----------+-----------------------+---------------+-------+-----+---------+---------+--------
   1 |  16.9568897845004 |  3.64613008870454 | 23.33542 | 21.95143 | 20.48149 |   19.603 | 19.13094 |   7712 |       6 |      442 | 4.855016555329904e+18 |     0.5062369 |  4312 |  55511 |      49 | 0.98686 | GALAXY
   2 |  240.063240247767 |  6.13413059813973 | 17.86033 | 16.79228 | 16.43001 | 16.30923 | 16.25873 |   3894 |       1 |      243 | 2.4489280322708705e+18 | 0.0003448142 |  2175 |  54612 |     348 | 0.990419 | STAR
   3 |  30.887222067625  |  1.18870964120799 | 18.18911 | 16.89469 | 16.42161 | 16.24627 | 16.18549 |   7717 |       1 |      536 | 8.255357438959835e+18 | 4.085216e-06  |  7332 |  56683 |      94 | 0.997588 | STAR
   4 |  247.594400505002 |  10.8877797153666 | 24.99961 | 21.71203 | 21.47148 | 21.30532 | 21.29109 |   5323 |       1 |      134 | 4.577998722756271e+18 | -0.0002914838 |  4066 |  55444 |      32 | 0.997667 | STAR
   5 |  18.8964507920807 | -5.26133022886992 | 23.76648 | 21.79737 | 20.69543 | 20.23403 | 19.97464 |   7881 |       3 |      148 | 8.91047176642785e+18  | -0.0001361561 |  7914 |  57331 |      36 | 0.996044 | STAR
   6 |  182.713733094955 |  51.3758050594777 | 22.44608 | 21.68444 | 20.24292 | 19.41423 | 19.08227 |   2830 |       1 |      411 | 7.516725588574623e+18 |     0.5026683 |  6676 |  56389 |      79 | 0.984373 | GALAXY
   7 |  150.089423193165 |  39.4670880748061 | 18.96441 | 17.82906 | 17.31429 | 16.99891 | 16.85583 |   3560 |       4 |      278 | 1.5267956411104236e+18 |  0.06366445 |  1356 |  53033 |      27 | 0.996164 | GALAXY
   8 |  189.510984338851 |  58.7411197772507 | 21.37376 | 20.80187 | 20.84925 | 21.13449 | 20.34689 |   2243 |       1 |      353 | 7.696817897528907e+18 |     0.7936153 |  6836 |  56443 |      60 | 0.957787 | QSO
   9 |  37.7138728560977 | -0.525138228146508 | 20.77988 | 19.54618 | 19.16687 | 18.89438 | 18.64286 |   2700 |       2 |      117 | 1.7553283123029217e+18 |     0.1060118 |  1559 |  53271 |      18 | 0.993892 | GALAXY
  10 |  201.074980072746 |  28.7699058867715 | 25.05349 | 22.23362 |  20.8122 | 19.69488 | 19.28336 |   4649 |       3 |      120 | 7.306035245308205e+18 |      0.567082 |  6489 |  56329 |      25 | 0.993856 | GALAXY
  11 |  151.83091832672  |  19.8108624669417 | 24.04443 | 22.48608 | 20.59701 | 19.50985 | 19.00457 |   5183 |       5 |      142 | 6.622787444780849e+18 |     0.5475619 |  5882 |  56029 |      88 | 0.998885 | GALAXY
```

SELECT * FROM ml_predict(...);

```
adult=# SELECT * from  ml_predict('astra3.cbm'
'astra3');
 id |      predict       | class
----+--------------------+--------
  0 | 0.9868595777513302 | GALAXY
  1 | 0.9904188657285139 | STAR
  2 | 0.9975875623929414 | STAR
  3 | 0.9976669380943318 | STAR
  4 | 0.9960439244920889 | STAR
  5 | 0.9843734017027631 | GALAXY
  6 | 0.9961635567874662 | GALAXY
  7 | 0.9577871819302538 | QSO
  8 | 0.9938922568658763 | GALAXY
  9 | 0.9938564131331261 | GALAXY
```

SELECT … FROM ml_predict_query(…)



Информация о модели: ml_info()

```
adult=# SELECT ml_info ('astra3.cbm');
                                                                                ml_info
-------------------------------------------------------------------------------------------------------------------------------
 dimension:3 numeric features:12 categorial features:0 modelType "MultiClass"+
 fieldName:alpha,delta,u,g,r,i,z,cam_col,redshift,plate,MJD,fiber_ID
(1 row)

adult=#
```

- Размерность результата
- Кол-во параметров
- Тип модели
- Имена полей

Информация о json модели: ml_json_info(<filename>)

```
adult=# SELECT * from ml_json_info('titanic.json');
   feature   | type
-------------+-------
 Pclass      | text
 PassengerId | text
 Cabin       | text
 Embarked    | text
 Ticket      | text
 Age         | float
 Name        | text
 Parch       | text
 SibSp       | text
 Sex         | text
 Fare        | float
(11 rows)
```



Информация о моделях

```
adult=# SELECT ml_info ('astra3.cbm');
                                   ml_info
-----------------------------------------------------------------------------
 dimension:3 numeric features:12 categorial features:0 modelType "MultiClass"+
 fieldName:alpha,delta,u,g,r,i,z,cam_col,redshift,plate,MJD,fiber_ID
(1 row)

adult=# SELECT ml_info ('titanic.cbm');
                                            ml_info
-------------------------------------------------------------------------------------------
 dimension:1 numeric features:2 categorial features:9 modelType "Accuracy"          +
 fieldName:PassengerId,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
(1 row)

adult=# SELECT ml_info('boston.cbm');
                                ml_info
-----------------------------------------------------------------------
 dimension:1 numeric features:13 categorial features:0 modelType "RMSE"  +
 fieldName:crim,zn,indus,chas,nox,rm,age,dis,rad,tax,ptratio,black,lstat
(1 row)
```

Пример Binary classification (Titanic,Adult)

```
adult=# SELECT index name,predict,class survived
  FROM ml_predict('titanic.cbm', 'titanic', '{name,passenger_id,pclass,sex,sibsp
,parch,ticket,cabin,embarked }', 'name') limit 10;
                    name                     |       predict        | survived
---------------------------------------------+----------------------+----------
 Kelly, Mr. James                            |  0.14261550408912857 | 0
 Wilkes, Mrs. James (Ellen Needs)            |   0.3109161066836635 | 0
 Myles, Mr. Thomas Francis                   |  0.08371776629817723 | 0
 Wirz, Mr. Albert                            |  0.12532120732529184 | 0
 Hirvonen, Mrs. Alexander (Helga E Lindqvist) |   0.3971202544185592 | 0
 Svensson, Mr. Johan Cervin                  |  0.11295647623974053 | 0
 Connolly, Miss. Kate                        |   0.6436970182793165 | 1
 Caldwell, Mr. Albert Francis                |  0.25376107990544644 | 0
 Abrahim, Mrs. Joseph (Sophie Halaut Easu)   |   0.6627874954602727 | 1
 Davies, Mr. John Samuel                     | 0.037292516669765206 | 0
(10 rows)
```

```
 FROM ml_predict ('adult.cbm',   'ad
s,education,marital_status, occupati
,race,sex,native country}') limit 10;
 index |       predict        | class
-------+----------------------+-------
 0     |   0.00266113320832485 | <=50K
 1     |   0.22690399673598352 | <=50K
 2     |   0.32114566576837716 | <=50K
 3     |    0.9742984083617117 | >50K
 4     |  0.009521778384767518 | <=50K
 5     |  0.004760746254986656 | <=50K
 6     |  0.019408350955775772 | <=50K
 7     |    0.5607339306273282 | >50K
 8     | 0.0035716251188914077 | <=50K
 9     |   0.09203603749238666 | <=50K
(10 rows)
```

Пример Regression (Boston)

```
adult=#   SELECT * from ml_cat_predict ('boston.cbm',  'boston2');
 row_num |      predict       | class
---------+--------------------+-------
       0 |  24.99820280685380 |
       1 | 20.664358727562394 |
       2 |  33.67737911788664 |
       3 |  34.28900239364565 |
       4 |  34.61570849423551 |
       5 | 27.968317495475695 |
       6 |  21.68218578618033 |
```

Пример Multi classification (Pulsar stars)

```
adult=# SELECT * from  ml_predict('astra3.cbm'
'astra3');
 id  |      predict       | class
-----+--------------------+--------
   0 | 0.9868595777513302 | GALAXY
   1 | 0.9904188657285139 | STAR
   2 | 0.9975875623929414 | STAR
   3 | 0.9976669380943318 | STAR
   4 | 0.9960439244920889 | STAR
   5 | 0.9843734017027631 | GALAXY
   6 | 0.9961635567874662 | GALAXY
   7 | 0.9577871819302538 | QSO
   8 | 0.9938922568658763 | GALAXY
   9 | 0.9938564131331261 | GALAXY
```

```
astra=# SELECT class, predict, index as name  FROM  ml_predict('exoplanets_
model_t.cbm','exoplanets_train_model_by_t' , '{spectr,type_t}', 'name');
   class    |       predict       |           name
------------+---------------------+--------------------------
 hot        |  0.9836827156493978 | Kepler-89 d
 hot        |  0.5973195076100603 | HD 28254 b
 ice        |  0.9732892668460026 | WASP-28 b
 hot        |  0.9989161355601188 | WASP-84 b
 warm       |  0.6641208386682923 | HD 106252 b
 ice        |  0.9940189501044985 | Kepler-8 b
 ice        |    0.95190596023499 | WASP-57 b
 hot        |  0.7579202916405879 | HD 173416 b
 hot        |  0.9981612189077735 | WASP-29 b
 ice        |  0.9245818348486754 | HD 3167 b
 hot        |   0.976449070282651 | K2-240 b
 ice        |  0.9949239070889919 | NGTS-24 b
 hot        |  0.9981352494125506 | TOI-421 b
 very hot   |  0.9609388933704017 | KELT-7 b
 hot        |  0.7460383200300617 | K2-16 c
 hot        |  0.5489876769123071 | HD 147018 c
 warm       |  0.6157999667711632 | HD 202206 c
 hot        |  0.8091935501883233 | HD 5319 b
```

PostgreSQL и ClickHouse (Titanic)









```
postgres@notebook-sasha: /usr/local/pgsql

adult=# SELECT * FROM  ml_predict ('titanic.cbm', 'titanic','{name,passen
ger_id,pclass,sex,sibsp,parch,ticket,cabin,embarked }', 'passenger_id') L
IMIT 10;
 index |       predict        | class
-------+----------------------+-------
 892   |  0.14261550408912857 | 0
 893   |   0.3109161066836635 | 0
 894   |  0.08371776629817723 | 0
 895   |  0.12532120732529184 | 0
 896   |   0.3971202544185592 | 0
 897   |  0.11295647623974053 | 0
 898   |   0.6436970182793165 | 1
 899   |  0.25376107990544644 | 0
 900   |   0.6627874954602727 | 1
 901   | 0.037292516669765206 | 0
(10 rows)

adult=#
```

Множественная классификация не поддерживается

```
akalend@notebook-sasha: ~

notebook-sasha :) SELECT PassengerId, catboostEvaluate('/tmp/titanic.cb
m',
 PassengerId, Pclass, Name, Sex, Age, SibSp,
 Parch, Ticket,
 Fare, Cabin, Embarked) AS prediction
FROM titanic
LIMIT 10

SELECT
    PassengerId,
    catboostEvaluate('/tmp/titanic.cbm', PassengerId, Pclass, Name, Sex
, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked) AS prediction
FROM titanic
LIMIT 10

Query id: cb0c6ac1-363e-4f4d-a172-54c06b6659b8

┌─PassengerId─┬──────────prediction─┐
│         892 │  -1.993659010153981 │
│         893 │   -2.15638521773551 │
│         894 │ -2.0175711183142973 │
│         895 │   -2.15638521773551 │
│         896 │ -2.0847733562958295 │
│         897 │   -2.15638521773551 │
│         898 │  -1.993659010153981 │
│         899 │  -2.108685464456145 │
│         900 │ -1.8606454378897987 │
│         901 │   -2.15638521773551 │
└─────────────┴─────────────────────┘

← Progress: 418.00 rows, 50.62 KB (33.25 thousand rows/s., 4.03 MB/s.)
↖ Progress: 418.00 rows, 50.62 KB (33.25 thousand rows/s., 4.03 MB/s.)

10 rows in set. Elapsed: 0.013 sec.

notebook-sasha :)
```

pg_ml и PostgesML (Titanic)







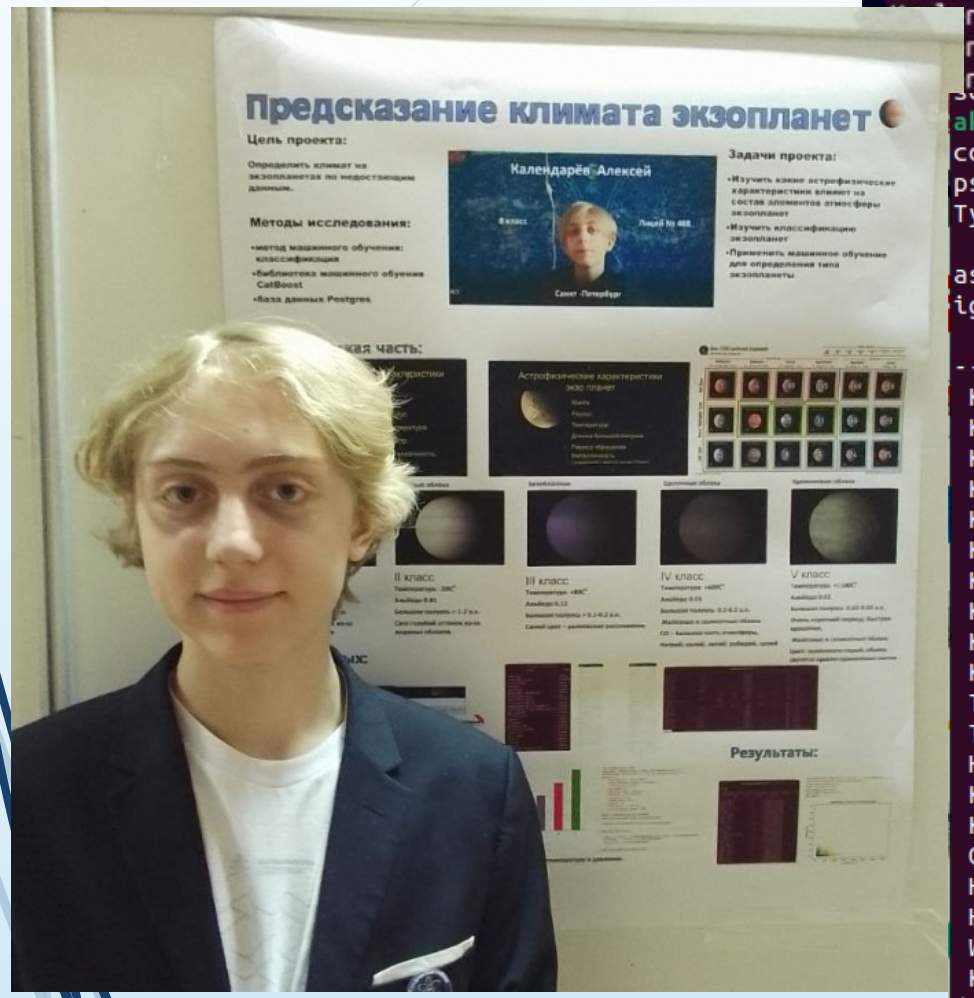
XGBoost

```
                    Table "public.titanic"
   Column    |       Type        | Collation |
-------------+-------------------+-----------+
id           | integer           |           |
passenger_id | integer           |           |
pclass       | integer           |           |
name         | text              |           |
sex          | text              |           |
age          | double precision  |           |
sibsp        | integer           |           |
parch        | integer           |           |
ticket       | text              |           |
fare         | double precision  |           |
cabin        | text              |           |
embarked     | character(1)      |           |
res          | boolean           |           |
```

On hot coding

```
postgresml=# \d titanic
                    Table "public.titanic"
     Column     |       Type        | Collation
----------------+-------------------+-----------
 index          | bigint            |
 unnamed:0      | bigint            |
 pclass         | bigint            |
 name           | text              |
 sib_sp         | bigint            |
 parch          | bigint            |
 ticket         | text              |
 fare           | double precision  |
 sex_male       | bigint            |
 nulls_1        | bigint            |
 nulls_2        | bigint            |
 cabin_mapped_1 | bigint            |
 cabin_mapped_2 | bigint            |
 cabin_mapped_3 | bigint            |
 cabin_mapped_4 | bigint            |
 cabin_mapped_5 | bigint            |
 cabin_mapped_6 | bigint            |
 cabin_mapped_7 | bigint            |
 cabin_mapped_8 | bigint            |
 embarked_q     | bigint            |
 embarked_s     | bigint            |
 survived       | bigint            |
Indexes:
    "ix_titanic_index" btree (index)
```

Использование




```
        name          | temperature |      a      |   mass    |      predict       |  cla
----------------------+-------------+-------------+-----------+--------------------+------
Kepler-480 b          |             |             |           |  0.770669842967865 | hot
r-783 b               |             |             |           |  0.411819276884600 | hot
r-783 c               |             |             |           | 0.4184813168285938 | hot
r-977 b               |             |             |           | 0.3587698343057905 | ice
                                                                          909839040 | hot
                                                                          686570063 | ice
                                                                          873988182 | ice
                                                                          105520188 | hot
                                                                          046128835 | hot
                                                                         9678291446 | hot
                                                                          349259450 | hot
                                                                          058915060 | hot
                                                                          286385844 | hot
                                                                          168947742 | ice
                                                                          124962224 | ice
                                                                          564895501 | ice
                                                                          491205212 | hot
                                                                          859991852 | hot
                                                                          449304990 | ice
                                                                          449304990 | ice
                                                                          449304990 | ice
                                                                          281845864 | ice
                                                                          739244085 | hot
                                                                          574365155 | hot
                                                                          346869291 | ice
                                                                          644797055 | hot
                                                                          552057695 | hot
                                                                          012489300 | hot
                                                                          354707272 | ice
akalend@notebook-sasha:~/stars$ ./showDb.sh
could not change directory to "/home/akalend/stars": Отказано в доступе
psql (15.1)
Type "help" for help.

astra=# select index as name, class, predict  from  ml_predict_query('gas_g
igants.cbm', 'select * from gigants', '{spectr}', 'name' ) limit 20;
         name          | class |      predict
-----------------------+-------+--------------------
 Kepler-480 b          | V     | 0.9951641235691848
 Kepler-783 b          | V     | 0.9950443609217626
 Kepler-783 c          | V     | 0.9938664778290038
 Kepler-977 b          | V     | 0.9824994611618907
 Kepler-1127 b         | V     | 0.9948356079238702
 K2-288 B b            | II    | 0.9922933057141604
 Kepler-705 b          | V     | 0.9799897906367515
 Kepler-1157 b         | V     | 0.9935370499434217
 Kepler-1607 b         | V     | 0.9881878161630462
 Kepler-141 c          | V     |  0.994220718248471
 TOI-813 b             | III   |  0.999253029786622
 TOI-1130 c            | III   | 0.9987479404592593
 HD 122562 b           | V     | 0.9653546557125285
 Kepler-1704 b         | III   | 0.9944261385535857
 K2-380 b              | V     | 0.9897188942162424
 OGLE-2005-BLG-390L b  | V     | 0.9540403386261808
 HD 28254 b            | II    |  0.997189343039423
 HD 116029 c           | V     | 0.9889323948885699
 WASP-28 b             | V     | 0.9997184696101044
 Kepler-141 b          | V     |  0.995368415496726
(20 rows)

astra=#
```

Перспективы

- Внедрение обучения через Python
- Создание нового модуля pg_catboost
- Внедрить синтаксис:

```
CREATE MODEL <name>
        TYPE <type> OPTIONS (. . .)
         AS (<query>)
 PREDICT <name> WITH  <options>
         AS <query>
```

Спасибо за внимание

Приглашаются желающие в проект

Вопросы?





@akalend



http://github.com/akalend/pg_ml